

CSE310: Compiler Sessional, January 2026 Term

ICG Online Assignment Practice

Built-in swap Statement

Time: 40 Mins

Total Marks: 10

Problem

Extend the grammar to support a built-in `swap` statement that exchanges the values of two variables in memory without requiring a temporary variable in the source code.

```
int a, b; a = 1; b = 2; swap(a, b);
```

Task

- Add a new statement rule to the grammar for `swap(ID, ID);`.
- Generate the assembly code to load both variable values into registers (e.g., `EAX` and `EDX`) and then move them into each other's respective memory locations.

Mark Distribution

- Extending the grammar for the `swap` statement. (4)
- Fetching and swapping memory contents correctly in assembly. (6)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)
<pre>int main() { int a, b; a = 10; b = 20; swap(a, b); println(a); println(b); return 0; }</pre>	<pre>main: PUSH EBP MOV EBP, ESP SUB ESP, 8 MOV EAX, 10 MOV [EBP-4], EAX MOV EAX, 20 MOV [EBP-8], EAX MOV EAX, [EBP-4] MOV EDX, [EBP-8] MOV [EBP-8], EAX MOV [EBP-4], EDX MOV EAX, [EBP-4] CALL print_number MOV EAX, [EBP-8] CALL print_number MOV EAX, 0 JMP .L1 .L1: ADD ESP, 8 POP EBP RET</pre>

Expected Output	20 10
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